

Antaral — architecture

A digital twin for the gap between dialysis sessions: patient twin → risk layer → unit twin → doctor's dashboard

1 DATA STREAMS

synthetic, sandbox-compliant

Static EHR

Age, dry weight, diagnosis
Drugs: ACE-i/ARB, beta-blocker
Residual urine, bicarbonate
Dialysis pattern (2x or 3x / week)

Wearables & home devices

Smartwatch single-lead ECG (2x/day)
Smart-scale morning weight
Steps, resting HR, HRV, sleep, BP
Food diary: coconut water, banana...

Dialysis unit records

Sessions: start, length, pre-weight
Potassium lab (~monthly)

Simulator (ground truth)

Hidden physiology, illness, unlogged food. Used only for evaluation.

2 PATIENT TWIN

one per patient, updated hourly

Mechanistic potassium model + Kalman filter

$$dK/dt = (\text{absorbed intake} - \text{renal} + \text{release}) / V$$

Personalised from the EHR
Dialysis: decay to 2.0 bath, then rebound
State: potassium K and personal drift b

ECG potassium sensor

Median beat: T height, width, slope, PR, QRS, QT vs own
post-dialysis beat → LightGBM
K estimate +/- sd (MAE 0.5)

What-if forecaster (Monte Carlo)

Sample (K, b), roll forward with noise
Scenarios: session time, food, binder, diet call → $P(K \geq 6 \text{ before session})$
and danger-hours over next two gaps

3 DECISIONS

every change needs a clinician's approval

Risk layer: LightGBM + calibration

Twin forecast + EHR + wearables + labs
→ calibrated risk of $K \geq 6.0$ before session
SHAP: why this patient, why tonight
Test AUROC 0.966 (last lab: 0.899)

Unit twin: nightly scheduler (MILP)

Tomorrow's chairs at 9 PM: maximise danger-hours avoided, same sessions/week, chairs, min/max gaps; stability penalty
Riskiest patients → earliest shift
Sim: ~15% fewer arrive with $K \geq 6$

Doctor's dashboard (Streamlit)

Unit tonight: suggestions + chair board
Patient twin: potassium trajectory, forecast fan, ECG vs baseline, food log, reasons
What-if panel for the clinician

EHR priors

food, weight

ECG beats

sessions, labs

K +/- sd

belief

forecast

what-if